

Fig. 5: Actual-size PCB layout of the transmitter circuit

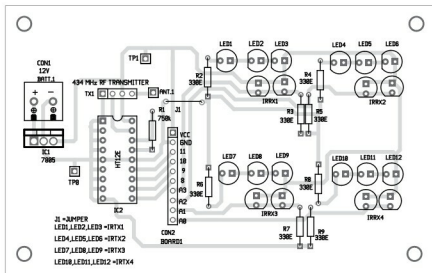


Fig. 6: Component layout for the PCB of transmitter circuit

sends signal to Arduino board. The signal received is first encoded using HT12E encoder and transmitted via an RF link module (TX1). The information is transmitted to the receiver module via RF link.

Receiver. These transmitted signals are received at the receiver end through an RF receiver module (RX1) and decoded via HT12D decoder. These signals are then transferred to AT89C2051 MCU on the receiving end to drive the robot.

Arduino Uno board. Arduino Uno is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is an MCU board based on ATmega328P. It has 14 digital input/output (I/O) pins (of which six can be used as pulse-width modulation outputs), six analogue inputs, a 16MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.

It contains everything needed to support the MCU; simply connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. Digital pins 8, 9, 10 and 11 of Arduino Uno are connected to pins 10, 11, 12 and 13 of HT12E encoder IC, respectively.

AT89C2051. AT89C2051 is an 8-bit MCU with 2k bytes flash, 128 bytes of RAM, 15 I/O lines, two 16-bit timers/counters and 5-vector two-level interrupt architecture. It receives decoded data from the RF receiver module through HT12D and instructs L293D (IC6) to drive the motor according to the signal bits received as shown in Table II.

Software

There are two software codes: one for the transmitter circuit using Arduino Uno and another for the receiver circuit comprising AT89C2051 MCU.

Arduino sketch (software) comprises the following function modules:

TABLE II

Signal Bits Transmitted and Received

Arduino output via digital pin (8, 9, 10, 11)	AT89C2051 MCU output via Port 3	Movement of the robot
1000	00001001b	Forward
0100	00000110b	Backward
0010	00001000b	Left
0001	00000001b	Right

as shown in Fig. 4. The circuit comprises some components that are described below:

Transmitter. The IR sensor module works on the principle of transmission of IR rays by IR LED

(IRTX1). Reflected IR ray from the object is received by the photodiode (IRRX1) that turns on the circuit. The remaining three IR modules operate similarly.

In this project, outputs from the four pairs of IR sensors are given to analogue pins A0, A1, A2 and A3 of Arduino Uno board. Each IR pair has unique information as listed in Table I. When you place your hands on a particular pair of an IR module, only that module is turned on and